Technical Test data visualisation in Python

This technical test consists of a few Python programming exercises aimed at:

1. Loading and plotting time series of aquifer temperature and groundwater levels.
2. Displaying sensor locations on a map.

You are free to use any tools you find useful to complete this test. However, the evaluation will focus primarily on your ability to explain and justify your code, rather than solely on the results it produces. The results themselves remain important.

Part 1. Plotting Time Series

The `piezo_data` folder contains three files (`pzps1.csv`, `pzps3.csv`, and `pzps4.csv`) corresponding to measurements collected by Diver sensors installed in monitoring wells (piezometers).

The measured variables are:

- Groundwater level (*Hydraulic Head [mNGF]*).
- Aquifer temperature (*Temperature [°C]*).
- Electrical conductivity (*Conductivity [mS/cm]*).

Exercises:

- Load the data from the `pzps1.csv`, `pzps3.csv`, and `pzps4.csv` files using Python.
- Calculate the warming trends for these piezometers using the `trend_detection_time` function provided in the `trends.py` file.
- Using the Plotly Python library, create a figure for each of the three sensors displaying:
    1. Groundwater level.
    2. Temperature.
    3. A straight line representing the warming trend.

Both variables (temperature and groundwater level) must be displayed on the same figure, using two separate y-axes (one on the left and one on the right).

The warming trend, expressed in degrees per decade (°C/decade), must be displayed in the figure title.

Part 2. Displaying Monitoring Stations on a Map

The `GIS` folder contains shapefiles storing geographical data, namely:

- The boundaries of the Orgeval catchment area (`limites_avenelles.shp`).
- The hydrographic network (`reseau_hydraugraphique.shp`).
- The metadata associated with the sensors (`stations_water_temp_piezo.shp`).

The latter contains, in particular:

- The geographical coordinates of the sensors (column `geometry`).
- The sensor names (column `LIBELLE`).
- The instrument type (river sensor or piezometer, column `TYPE_INSTR`).

The `mapping.py` file contains a function called `map_sensors_orgeval` that needs to be completed. Its purpose is to display the sensors on a map. The function already plots the hydrographic network and the catchment boundaries.

Exercises:

- Complete the function so that it also displays the location of the sensors on the map.
- Display the name of each sensor next to its corresponding point.
- Use different colors depending on the type of sensor (piezometer or river sensor).
- Load the data and use the `map_sensors_orgeval` function to display the results on the map.

You may select a subset of sensors to improve the readability of the map, as some sensors are located very close to one another.